

WASP-2b

R-1174

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_200712 · created 2026-07-31T03:38:50+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459042.7702 +/- 0.0062 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.046
Transit depth (Rp/Rs)^2	2.35 +/- 1.4 [%]
Orbital Inclination (inc)	85.62 +/- 2.89
Airmass coefficient 1 (a1)	0.86 +/- 0.54
Airmass coefficient 2 (a2)	0.11 +/- 0.56
Scatter in the residuals of the lightcurve fit is	63.85 %
Best Comparison Star	None
Optimal Aperture	2.05
Optimal Annulus	4.5
Transit Duration (day)	0.072 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.046 vs 0.1326 (deviation 0.41σ)
Duration fit vs published	0.072 d vs 0.0737 d (deviation 0.05σ)
Mid-time O–C	-14.6 min
Depth SNR	3.1
Residual scatter	63.85 %

Light curve

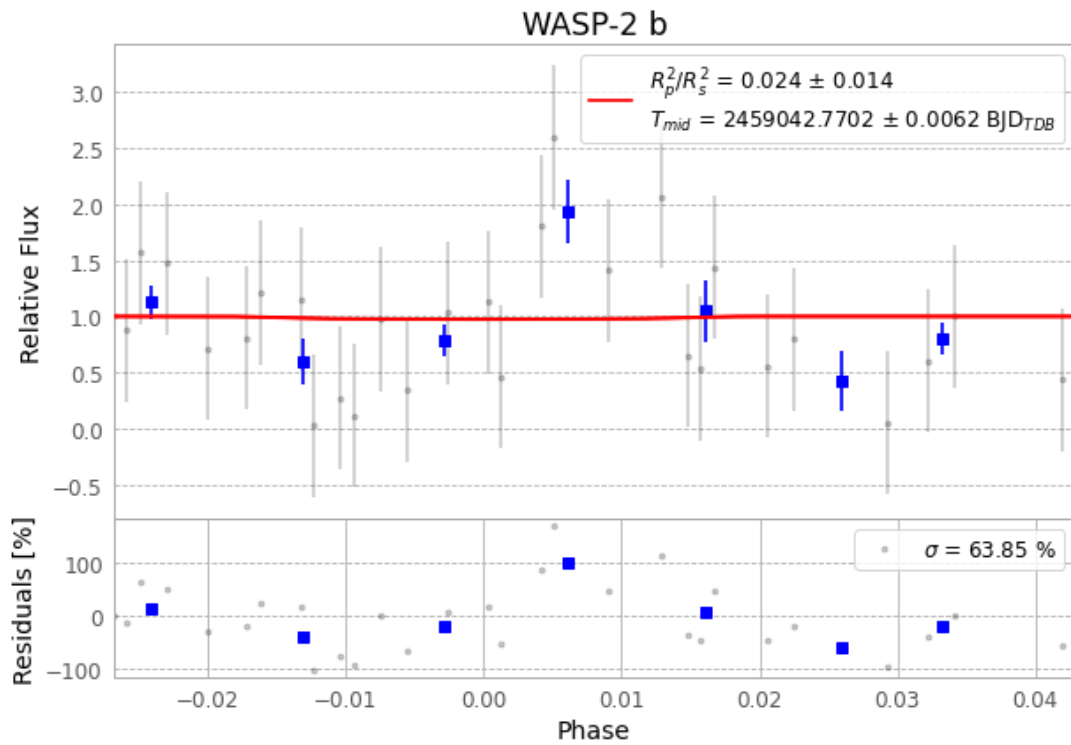


Figure 1 — FinalLightCurve_WASP-2 b_2020-07-11.png (SHA-256 b5a3282775f12359...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [311, 126], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e84c1cf8882adc48...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-2200712045112.FITS → WASP-2200712083612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-200712.log (b1552f5ede02...), FinalLightCurve_WASP-2 b_2020-07-11.png (b5a3282775f1...), FinalLightCurve_WASP-2 b_2020-07-11.csv (7f271fe7c915...)

Compute resources

Wall time 180.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 03:38Z.